

SEMESTER END EXAMINATIONS – AUGUST 2024

Program	: B.E. – Information Science and Engineering	Semester	: IV
Course Name	: Microcontrollers	Max. Marks	: 100
Course Code	: IS42 / IS46	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Discuss the four major design rules that are taken into considerations in implementation of RISC systems. CO1 (08)
 - With an appropriate diagram, discuss the PIN configuration of an MCU that shows the set of pins associated with each of the peripherals and its functionalities. CO1 (08)
 - Why does the clock use PLL? Justify. CO1 (04)
- Discuss the characteristics of an embedded systems, also describe the applications of embedded systems. CO1 (08)
 - Compare the RISC and CISC type of systems. CO1 (06)
 - What is the relevance of duty cycle? Define the duty cycle to showcase for the 50% of its usage. CO1 (06)

UNIT - II

- Mention the positive aspects of I2C protocol. CO2 (07)
 - With a diagram, explain Serial peripheral interface (SPI) on-board protocol. CO2 (06)
 - With a neat diagram, show the classification of semiconductor memory types. Explain memory read and write cycle of SRAM chip. CO2 (07)
- For the following operations, find out which flags will be set, (if any) for a 16-bit processor? CO2 (08)
 - Add 0x6789 and 0x1234
 - Subtract 0xFE45 from 0x0978
 - Compare A and B where A = 0x78 and B = 0x78
 - Add 0xEF39 and 0xCD34.
 - Explain Load Store architecture used in RISC processors. CO2 (06)
 - Discuss aligned and misaligned data in ARM7 processor. CO2 (06)

UNIT - III

- Discuss on how the AMBA processor specifies the use of AHB & APB bus components. CO3 (08)
 - Find the output of the following instructions given that R3= 0xFF, R2=0x14. CO3 (06)
 - AND R4,R2,R3
 - ANDS R6,R2,R3
 - BIC R5,R2,R3.
 - Write an ALP to add the first 5 natural numbers, result to be stored in a register. CO3 (06)

6. a) Discuss with an appropriate diagram for the register set working towards various modes of an ARM 32 bit processor. CO3 (08)
b) Discuss the additional features of the ARM7 processor that are accepted in the embedded market. CO3 (06)
c) Write an assembly program to find the arithmetic progression of a given number. CO3 (06)

UNIT- IV

7. a) Write an Assembly language program to add first 10 odd numbers starting from 1. CO4 (06)
b) Discuss the following Assembler directives: CO4 (08)
i. AREA
ii. DCD, DCW
iii. ENTRY
iv. EQU
c) What is the content of the following register R6 after the execution of each of the following instructions, given that R6=0x44001100 and R2=0x 02. CO4 (06)
i. LDR R2, [R6, #0x0056]!
ii. STR R1, [R6, R2]!
iii. LDR R3, [R6, LSL #4]
8. a) Write an assembly language program to find 10 Fibonacci numbers excluding the first one. CO4 (10)
b) List the important features of LPC 214x series. CO4 (05)
c) Write a C program to display the string 'I LOVE ISE' in the serial window of UART1. CO4 (05)

UNIT- V

9. a) Draw a block diagram showing the mandatory functional blocks in the cortex-M0. CO5 (10)
b) Discuss how priority assignment is done for all interrupts happening for the internal, external and system exceptions. CO5 (10)
10. a) How has cortex-M0+ achieved the lower power levels than the cortex-M0? Justify. CO5 (10)
b) Discuss the memory model of cortex-M0. CO5 (10)
